

Overall scientific questions.

Introduction

- Overview of the ISM. How dust forms, why we use CO as a tracer. Why we might use other tracers.
- Jeans collapse a model for star formation.
- IMF / CMF. How the high mass end changes. SF across different environments.
- Quasistatic turbulent core fragmentation as a model for star formation.
- (Nico and Alessio) Literature on decoupling cores from clumps. We would like to know at what stage the gas decouples.
- CHIMPS paper and other extragalactic attempts at this sort of work.

The interstellar medium (ISM) is made up of primarily gas, as well as dust and cosmic rays. The ISM can be divided up based on temperature, ionisation and density, it consists of 5 main phases: the cold molecular medium (CMM), which is defined such that the gas is molecular with a temperature range of $10 - 30\text{ K}$ and a number density between $10^2 - 10^6\text{ cm}^{-3}$; the cold neutral medium (CNM), which is defined such that the gas is mostly atomic with a temperature range of $30 - 100\text{ K}$ and a number density between $1 - 100\text{ cm}^{-3}$; the warm neutral medium (WNM), which is defined such that the gas is mostly atomic with a temperature range of $6000 - 10000\text{ K}$ and number densities below 1 cm^{-3} ; the warm ionised medium (WIM), which is defined such that the gas is mostly ionised with a temperature range of $8000 - 10000\text{ K}$ and number densities around 0.4 cm^{-3} and the hot ionised medium (HIM), which is defined such that the gas is mostly ionised with a temperature range of $10^6 - 10^7\text{ K}$ and number densities below 10^{-2} cm^{-3} (see [Klessen and Glover 2016](#)). The focus of this work will be on the CMM.

Other than primordial elements, every atom formed inside stars. Therefore, initially these elements that make up the gas are hot and have not formed in their molecular state. To get to the temperature of the CMM, the gas must cool. A star ejects matter throughout its life as stellar

winds, as well as at the end of its life in different ways depending on its mass. As the gas moves away from its progenitor the number of ionising photons will decrease which will offset the balance between ionisation and recombination. At these higher temperatures the cooling from Lyman- α becomes important. The general form of the cooling rate of a two level system is shown below.

$$\Lambda_{ul} = \frac{g_u}{g_l} A_{ul} E_{ul} n_l e^{-E_{ul}/k_B T} \quad (1)$$

[Equation 1](#) is the rearranged form of 2.38 in [Tielens 2005](#). The formula comes about by assuming LTE and that the number density is much less than the critical density. g_u and g_l are statistical weights of the energy states, A_{ul} is the Einstein A coefficient of spontaneous emission, E_{ul} is the energy of the transition and n_l is the number density of the lower state.

$$\Lambda_{HI} = 7.3 \times 10^{-19} e^{-118400/T} n_e n_H \quad (2)$$

[Equation 2](#) appears as 2.70 in [Tielens 2005](#). Above 10^4 K the exponential term stops being negligible which causes the cooling rate to rise rapidly. This alongside the recombination to form atomic hydrogen is what facilitates the transition to the WNM. At higher metallicities the cooling is not dominated by Lyman- α but by other fine structure transitions from heavier elements. Once cooled, [Equation 2](#) shows a ceiling for the temperature in the WNM above which the cooling rate can rapidly exceed the different heating rates ([Tielens and Hollenbach 1985](#)). A different fine structure transition dominates the cooling in the WNM and facilitates the transition to the CNM, the CII transition. The energy of this transition is 92 K ([Hollenbach and McKee 1989](#)) meaning the exponent in [Equation 1](#) becomes $e^{-92/T}$. This means that below 100 K the cooling from CII decreases rapidly. On the other hand, the first rovibrational transition ($J = 1-0$) of CO emits a photon with a rest frequency of 115.27 GHz ([Winnewisser et al. 1997](#)), which gives an energy of 5.53 K , using $h\nu/k_B$. This allows the cooling rate of CO to dominate in this temperature regime and allow the gas to get to a temperature of around 10 K . Even though

CO is a more efficient coolant, CII seems to be a more dominant cooling mechanism at low metallicity as well as for diffuse gas with densities less than 10^3cm^{-3} (Glover and Clark 2012). This is partly because at low metallicity, the CO cannot be shielded by the dust and photodissociates by way of the ISRF. It is also important to note that Glover and Clark 2012 indicate that if the cloud is shielded from the effects of photoelectric heating then cooling from CII can allow the gas to get down to temperatures of 20K without the presense of CO.

Once there is gas at the typical temperatures of the CMM, molecular hydrogen can begin to form. Molecular hydrogen should form directly between two atoms of hydrogen but the Einstein A coefficient is too small for the process to allow for a photon to be emitted during their collision (Wakelam et al. 2017). This means that molecular hydrogen cannot form efficiently this way, therefore another mechanism is required to get the amount of molecular hydrogen expected. The process involves atomic hydrogen sticking to a dust grain and potentially colliding with another atom of hydrogen allowing for the recombination of molecular hydrogen. The energy produced by this is enough to break the bonds between the dust grain and the molecule of hydrogen, allowing for it to return to the CMM (Wakelam et al. 2017). This has been confirmed experimentally using various dust grain substitutes made of carbon or silicates. Katz et al. 1999 find that the peak efficiency of the recombination occurs below 20K for both types of grain. The formation of CO is very complex (see Glover et al. 2010) and can involve different pathways, the simplest of which is the formation from a carbon atom and a hydroxyl radical. Even though it is thought that the gas being in a molecular state is not necessary for star formation, the cloud being able to shield itself from the ISRF is paramount to the formation of both stars and molecular gas (Glover and Clark 2012). They find that a cloud of atomic gas is still capable of forming stars at the same rate as if molecular gas is able to form. This however does not change the fact that molecular hydrogen still traces star formation, it just may not trace all star formation.

Tracing molecular gas is important because it allows for the understanding of the steps that lead to star formation. CO is regarded as a good tracer of molecular gas for its abundance in molecular clouds and spontaneous low energy transitions, like the one described above. The

idea is to trace the more abundant molecular hydrogen, which is not directly possible due to its lack of emission at the typical temperatures of the CMM. The lowest possible rotational transition of molecular hydrogen is at an energy of 510K (Wakelam et al. 2017). CO has been the gold standard when it comes to tracing molecular hydrogen, but tracers like HCN or N_2H^+ have been found to trace the dense gas much more effectively than CO (Kauffmann et al. 2017). N_2H^+ is almost always being detected above column densities of 10^{22}cm^{-2} (Fehér et al. 2025), meaning it is much better at tracing the clumps where stars are likely to be forming.

Regions of the ISM that contain the CMM are also known as molecular clouds. These clouds are the densest constituents of the ISM, the collapse of which leads to star formation. For a cloud to collapse the gravitational force pulling the gas together must overcome that of the internal pressure of the gas. Hydrostatic equilibrium is the point at which both forces cancel one another out. This allows for the derivation of the mass above which a spherical system would collapse, known as the Jeans mass.

$$M_J = \frac{\pi^{3/2} c_s^3}{\rho_o^{1/2} G^{3/2}} \quad (3)$$

Equation 3 appears as the rearranged form of 9.23 in Stahler and Palla 2004 using the fact that $M_J \sim \rho_o \lambda^3$. c_s refers to the isothermal sound speed in the medium given by $\sqrt{k_B T / \mu m_p}$. This equation assumes that the system is isolated with no external forces, but the ISM is turbulent therefore an external pressure could be present. In this instance the external pressure would oppose the internal pressure of the gas, making it easier for the gas to collapse. This mass is known as the Bonner-Ebert mass.

$$M_{BE} = \frac{1.18 c_s^4}{P_o^{1/2} G^{3/2}} \quad (4)$$

Equation 4 appears as 9.16 in Stahler and Palla 2004. P_o is the external pressure on the cloud. There are several mechanisms that affect star formation that these equations do not account for. Magnetic fields oppose the gravitational collapse of clouds, slowing or stopping the collapse of clouds (Krasnopolsky and Gammie 2005). This allows more gas to be accreted before collapse leading to higher mass star formation. Stellar feedback strips the diffuse gas from the cloud slowing their growth but allowing the dense centres to continue to be star forming (Watkins et

al. 2025). It can also cause the destruction of the host cloud. Watkins et al. 2025 also found that stellar feedback likely breaks apart larger clouds into smaller clouds. Protostellar radiative feedback stops fragment formation causing most of the gas to be accreted onto a few objects leading to higher mass star formation (Krumholz, Klein, and McKee 2007). Cloud-cloud collisions between both atomic and molecular clouds facilitates the formation of massive molecular cloud cores, making them the dominant mechanism for forming high mass stars. Additionally, there is observational evidence that they are responsible for starbursts (e.g. the Antenna galaxies) (Fukui et al. 2021).

As a cloud collapses, the Jeans mass scales proportionally with temperature and density ($M_J \propto \rho^{-1/2} T^{3/2}$), as shown by Equation 3. During gravitational collapse the density always increases and providing the gas is isothermal, the Jeans mass decreases as the density increases (Ward-Thompson and Whitworth 2011). Therefore, sections of the collapsing cloud become Jeans unstable and begin gravitational collapse causing multiple collapsing regions. These collapsing regions can go on to fragment again. Eventually the density increases so much so heat is unable to radiate away and cooling becomes unable to keep the gas isothermal, therefore temperature increases. Assuming the collapsing cloud acts like a polytrope, it will follow the equa-

tion of state ($P = K\rho^\gamma$), where γ is the adiabatic index (Chieze 1987). If heat cannot radiate away then the collapse can be considered adiabatic, so the temperature is now a function of density ($T = K\rho^{\gamma-1}$). This now means that the Jeans mass only scales proportionally with density ($M_J \propto \rho^{(3/2)(\gamma-4/3)}$). In molecular clouds, the gas is primarily diatomic or monatomic which have 5 and 3 degrees of freedom, which gives them adiabatic indices of 5/3 and 7/5, respectively ($\gamma = 1 + 2/d.o.f.$). Both these indices result in a direct proportionality between the Jeans mass and density, causing fragmentation to cease.

The above picture of fragmentation is not complete as it does not take into account turbulence and magnetic fields. Magnetic fields as described above stop fragmentation and so the theory of ambipolar diffusion was introduced, which allows mass to move across field lines. Ambipolar diffusion

Project Work

- Motivations (link back to the lit review).
- Methods.
- Conclusions and what that might mean.
- Future work.

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